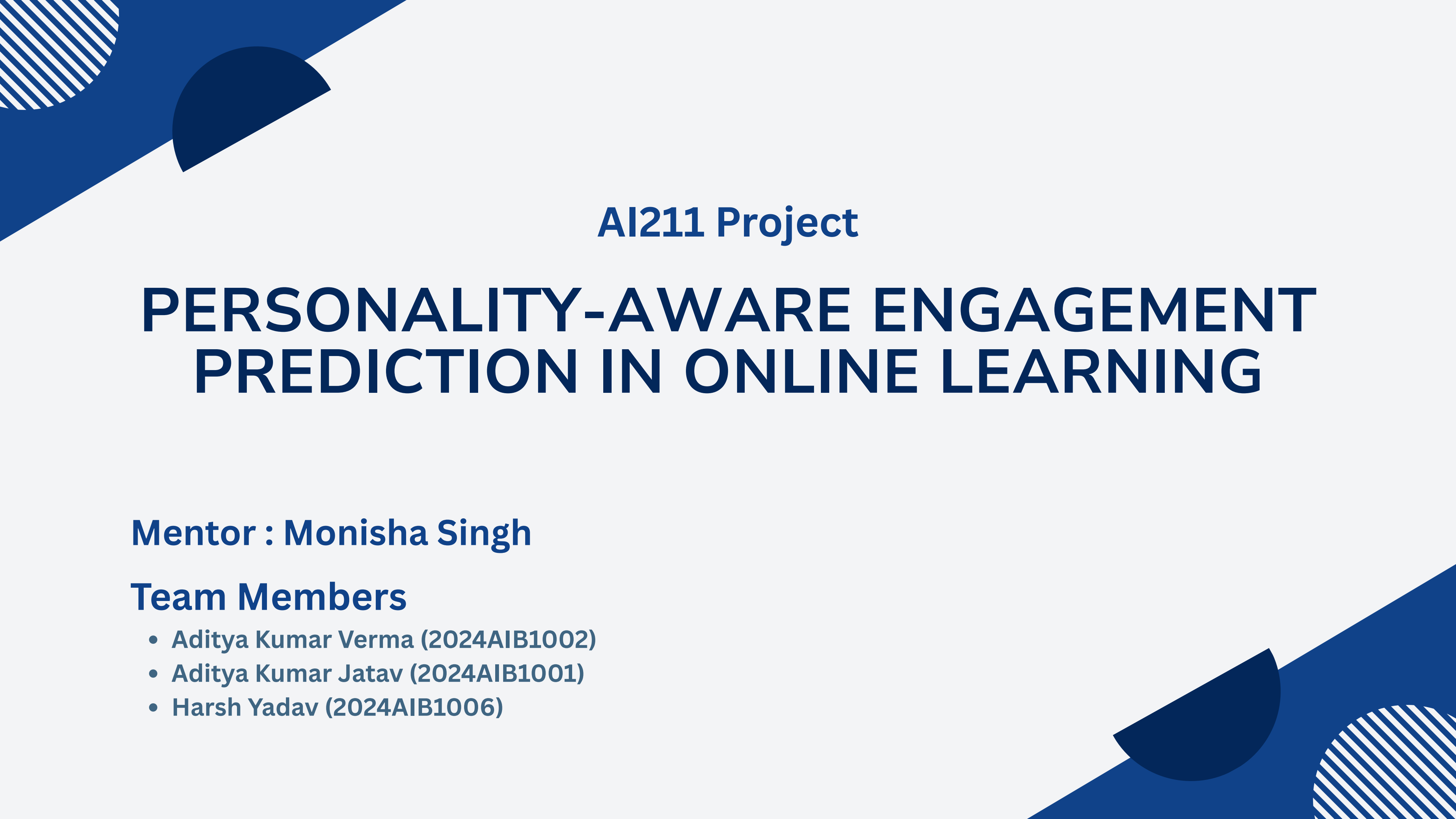




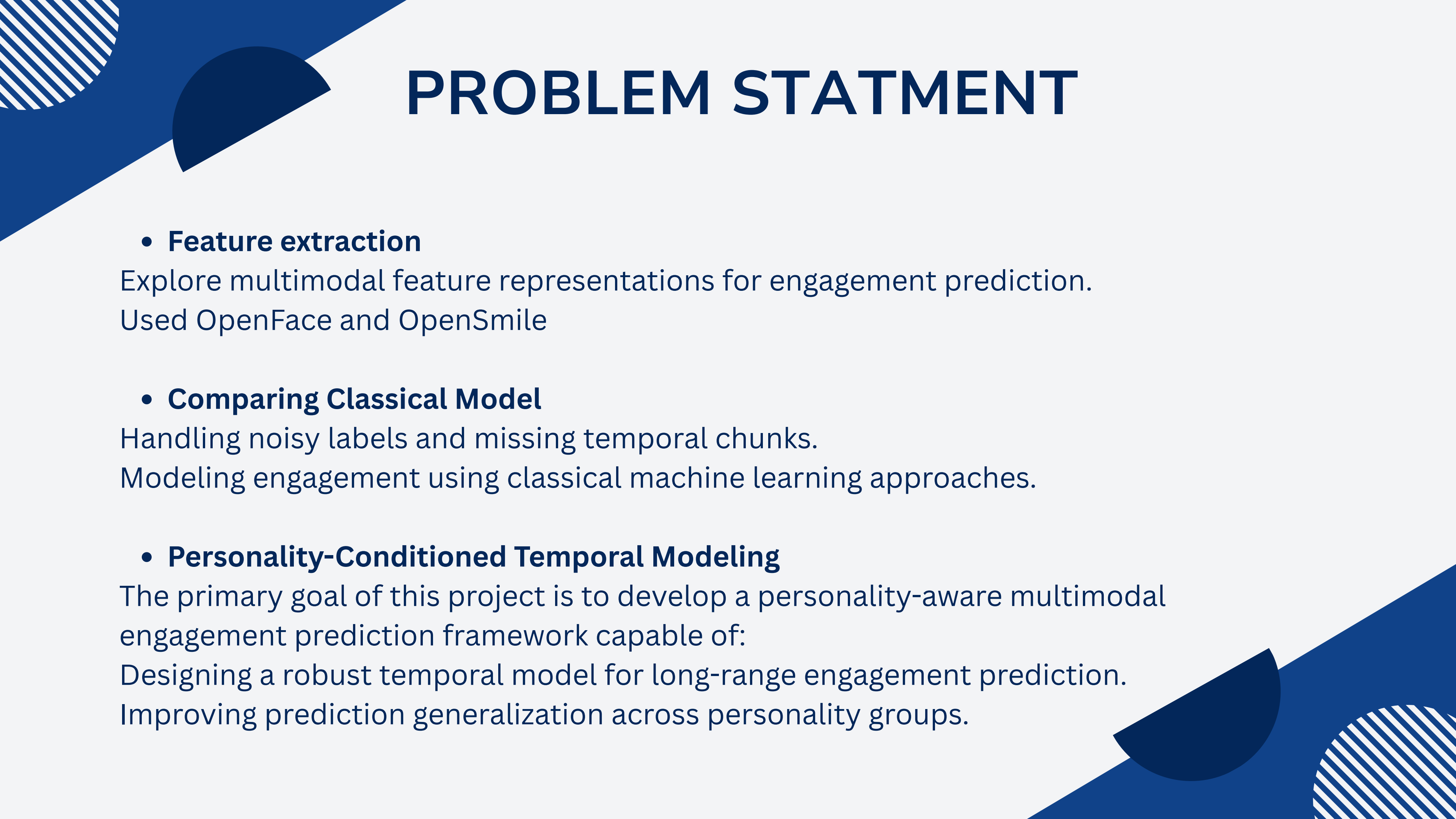
AI211 Project

PERSONALITY-AWARE ENGAGEMENT PREDICTION IN ONLINE LEARNING

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PROBLEM STATEMENT

- **Feature extraction**

Explore multimodal feature representations for engagement prediction.
Used OpenFace and OpenSmile

- **Comparing Classical Model**

Handling noisy labels and missing temporal chunks.
Modeling engagement using classical machine learning approaches.

- **Personality-Conditioned Temporal Modeling**

The primary goal of this project is to develop a personality-aware multimodal engagement prediction framework capable of:
Designing a robust temporal model for long-range engagement prediction.
Improving prediction generalization across personality groups.

MOTIVATION

- Understanding engagement is essential for evaluating and **improving online learning** outcomes.
- Standard models often evaluate short, isolated segments of behavior, which can **fail to capture long-range temporal dynamics** or the inherent noise in behavioral annotations.

Why include personality traits?

- **Engagement is highly subjective.** By integrating personality traits into long-range temporal modeling, the resulting system can offer a much more nuanced, individualized, and accurate assessment of user engagement.

METHODOLOGY

Feature Extraction

Open Face: It processes images, video streams, or webcam feeds to track and **quantify facial movements and expressions** over time.

Outputs : a highly detailed CSV file for each video, containing frame-by-frame data

Features

- Facial Action Units (AUs)
- Head Pose
- Eye Gaze
- Facial Landmarks

OpenSmile : Strips away the semantic meaning of the words (what is being said) and **focuses purely on the paralinguistics (how it is being said).**

Features

- Prosodic Features
- Spectral Features
- Voice Quality

OpenFace and OpenSmile operate at different sampling rates (e.g., video at 30 frames per second, but audio windowed every 10 milliseconds)
so we just went with Open Face feature for Task 2

METHODOLOGY

Temporal Information

- **Temporal aggregation:** summarize each time window using statistical descriptors (e.g., mean, variance, extrema) to capture overall behavioral patterns.
- **Dynamic feature engineering:** encode temporal change through derivatives, trends, or motion-based features reflecting behavioral dynamics.
- **Sliding-window representation:** treat short overlapping temporal segments as independent samples to preserve local temporal context.

PCA

```
#keep 90% of variance  
pca_variance_ratio = 0.90
```

```
Processing Temporal Aggregation  
Original shape: (1178, 999)  
Reduced shape:  (1178, 12)  
Components retained: 12
```

```
Processing Dynamic Features  
Original shape: (1178, 333)  
Reduced shape:  (1178, 8)  
Components retained: 8
```

```
Processing Sliding Window  
Original shape: (11780, 3330)  
Reduced shape:  (11780, 11)  
Components retained: 11
```


METHODOLOGY

Short-Context Classical Model

```
param_grids = {  
    "Logistic Regression": {  
        'C': [0.01, 0.1, 1, 10],  
        'solver': ['liblinear', 'lbfgs']  
    },  
    "SVM": {  
        'C': [0.1, 1, 10],  
        'gamma': ['scale', 'auto', 0.1],  
        'kernel': ['rbf']  
    },  
    "Random Forest": {  
        'n_estimators': [50, 100],  
        'max_depth': [None, 10, 20],  
        'min_samples_split': [2, 5]  
    },  
    "XGBoost": {  
        'learning_rate': [0.01, 0.1],  
        'max_depth': [3, 6],  
        'n_estimators': [100, 200]  
    },  
    "MLP": {  
        'hidden_layer_sizes': [(64, 32), (128, 64)],  
        'learning_rate_init': [0.001, 0.01]  
    }  
}
```

Evaluation Method	Model	Best Parameters	Train Accuracy	Train F1-Score	Test Accuracy	Test F1-Score
Temporal Aggregation PCA	Logistic Regression	C: 0.1, solver: 'lbfgs'	0.5276	0.5246	0.4958	0.4957
	SVM	C: 10, gamma: 'auto', kernel: 'rbf'	0.9851	0.9851	0.4873	0.4794
	Random Forest	max_depth: 10, min_samples_split: 5, n_estimators: 100	0.9735	0.9734	0.4873	0.4868
	XGBoost	learning_rate: 0.01, max_depth: 3, n_estimators: 100	0.6518	0.6516	0.5042	0.5027
	MLP	hidden_layer_sizes: (128, 64), learning_rate_init: 0.001	0.9745	0.9745	0.5	0.4987
Dynamic Features PCA	Logistic Regression	C: 0.01, solver: 'liblinear'	0.5382	0.5279	0.4534	0.438
	SVM	C: 1, gamma: 0.1, kernel: 'rbf'	0.6306	0.6257	0.4492	0.4388
	Random Forest	max_depth: 10, min_samples_split: 2, n_estimators: 100	0.9756	0.9756	0.4576	0.4576
	XGBoost	learning_rate: 0.01, max_depth: 3, n_estimators: 100	0.6582	0.6582	0.4449	0.4432
	MLP	hidden_layer_sizes: (128, 64), learning_rate_init: 0.01	0.6794	0.6691	0.5085	0.4981
Sliding Window PCA	Logistic Regression	C: 10, solver: 'liblinear'	0.5267	0.5266	0.5348	0.5348
	SVM	C: 10, gamma: 'auto', kernel: 'rbf'	0.9817	0.9817	0.8145	0.8124
	Random Forest	max_depth: None, min_samples_split: 2, n_estimators: 100	0.983	0.983	0.7992	0.7991
	XGBoost	learning_rate: 0.1, max_depth: 6, n_estimators: 200	0.9308	0.9308	0.7424	0.7423
	MLP	hidden_layer_sizes: (128, 64), learning_rate_init: 0.001	0.8989	0.8986	0.7525	0.7514

METHODOLOGY

Methodology: Personality-Conditioned Temporal Modeling

- **Personality Integration:** Learner and instructor personality traits (20D total) were combined and projected into a 128D latent representation, which initialized the Bi-LSTM hidden state to provide behavioral context.
- **Model Architecture:** A 2-layer Bidirectional LSTM (hidden size 128) processed temporal visual and acoustic sequences. Forward and backward hidden states were concatenated and passed through an MLP with ReLU activation to predict engagement scores.
- **Training & Robustness:** The model used the Adam optimizer with Huber Loss for resilience against noisy human annotations. Dropout (0.3) was applied between LSTM layers to reduce overfitting and improve generalization.

RESULT

FINAL SUMMARY: ALL MODELS ACROSS ALL TEMPORAL STRATEGIES

TEST ACCURACY

Temporal Strategy	dynamic features pca	sliding window pca	temporal aggregation pca
Model			
Logistic Regression	0.4534	0.5348	0.4958
MLP	0.5085	0.7525	0.5000
Random Forest	0.4576	0.7992	0.4873
SVM	0.4492	0.8145	0.4873
XGBoost	0.4449	0.7424	0.5042

TEST F1-SCORE

Temporal Strategy	dynamic features pca	sliding window pca	temporal aggregation pca
Model			
Logistic Regression	0.4380	0.5348	0.4957
MLP	0.4981	0.7514	0.4987
Random Forest	0.4576	0.7991	0.4868
SVM	0.4388	0.8124	0.4794
XGBoost	0.4432	0.7423	0.5027

```
PS D:\ai211_proh> python .\task3_evaluate
Using device: cuda
Matched 1178 video sequences
Evaluating model...
MAE: 0.4598
F1 Score: 0.6527
Accuracy: 0.6053
CCC: 0.1506
Introverts (n=510):
MAE: 0.4624
Accuracy: 0.6000
Extroverts (n=668):
MAE: 0.4579
Accuracy: 0.6093
PS D:\ai211_proh>
```

On the left is the output for the classical machine learning models with different temporal strategies, and on the right are the metrics for the LSTM. We can see that the LSTM achieved an accuracy of 60 percent, while the SVM achieved a lot higher accuracy of about 80 percent. Thus, we can say that SVM with sliding window was the best model overall.

CONCLUSION

In conclusion, we noticed that most classical ML models do not perform very well in the task of engagement prediction, with their accuracies being very close to random guessing. We saw that Sliding window was the best strategy to incorporate temporal data, with accuracies of many models jumping to over 70 percent in sliding windows case. We also noticed that the LSTM, which incorporated the instructor and student personality data, didnt perform as well as the best classical models, though it had the advantage of not being reliant on a good temporal flattening strategy and can also be used in online large scale settings and can work with missing data and noisy labels.



THANK
YOU